

Chapter 4 : Quantum Information Science

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In this chapter, we study basic concepts to prepare our journey to the quantum machine learning.

Topics include basic algorithms, quantum parallelism, teleportation, and associated programming in Qiskit and Cirq.

Part 1 : Qiskit and Cirq to construct quantum circuits and states, Von Neumann Entropy

Part 2 : Quantum Teleportation, Gate Scheduling, Quantum Parallelism and Function Evaluation, Deutsch's Algorithm

Part 1 : Quantum Circuits with Qiskit and Cirq , and Von Neumann Entropy

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Quantum states and Observables: A review

A **pure state** is a vector $|\psi\rangle$ in a Hilbert space $\mathbb{H}$.

In our case, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.

An **observable** A is a self-adjoint operator on $\mathbb{H}$.

We compute the mean value of A by $\langle A \rangle_\psi := \langle \psi | A \psi \rangle$.

The probability $P_\psi(\lambda)$ that for a quantum system in the pure state $|\psi\rangle \in \mathbb{H}$, a **measurement** of the observable yields the eigenvalue λ of A is given by

$$P_\psi(\lambda) = \|P_\lambda |\psi\rangle\|^2,$$

where P_λ is the projection onto the eigenspace $\text{Eig}(A, \lambda)$.

Quantum states and Observables: A review

Projection postulate : If a measurement of the observable A on a quantum mechanical system in the pure state $|\psi\rangle \in \mathbb{H}$ yields the eigenvalue λ , then

$$|\psi\rangle = \text{state before measurement} \rightarrow \frac{P_\lambda |\psi\rangle}{\|P_\lambda |\psi\rangle\|} = \text{state after measurement.}$$

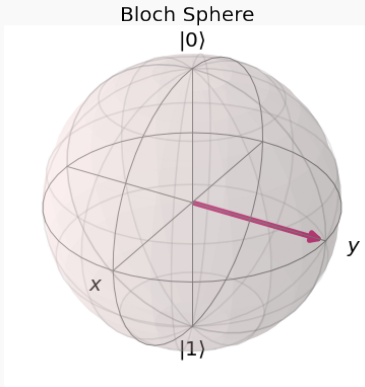
A **density operator** is an operator ρ acting on $\mathbb{H}$ which is self-adjoint, positive ($\rho \geq 0$), and has trace 1. Then, the quantum states described by density operators are called **mixed states**.

Indeed, we can write

$$\rho = \sum_j p_j |\psi_j\rangle \langle \psi_j|$$

Bloch representation of pure and mixed states

Since a qubit $|\psi\rangle \in \mathbb{HS}^1$, we can represent it on a sphere.



Bloch representation of pure and mixed states

To be more specific, $|\psi\rangle = a|0\rangle + b|1\rangle$ with $|a|^2 + |b|^2 = 1$ allows a parametrization

$$|\psi\rangle = e^{-i\frac{\phi}{2}} \cos \frac{\theta}{2} |0\rangle + e^{i\frac{\phi}{2}} \sin \frac{\theta}{2} |1\rangle.$$

For a density operator

$$\rho = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

using $\rho^* = \rho$ and $\text{tr}(\rho) = 1$, we get

$$\rho = \frac{1}{2}(1 + \mathbf{x} \cdot \boldsymbol{\sigma}) = \frac{1}{2} \begin{pmatrix} 1 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & 1 - x_3 \end{pmatrix},$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$.

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Getting started with python libraries

Qiskit : Library from IBM

Cirq : Library from Google

Getting started with Qiskit

Good introduction and guide for using the Qiskit.



Getting started with Qiskit

Anaconda prompt $\Rightarrow$ pip install qiskit

```
(base) C:\Users\admin>pip install qiskit
Requirement already satisfied: qiskit in c:\users\admin\anaconda3\lib\site-packages (0.35.0)
Requirement already satisfied: qiskit-ignis==0.7.0 in c:\users\admin\anaconda3\lib\site-packages (from qiskit) (0.7.0)
Requirement already satisfied: qiskit-aer==0.10.3 in c:\users\admin\anaconda3\lib\site-packages (from qiskit) (0.10.3)
Requirement already satisfied: qiskit-terra==0.20.0 in c:\users\admin\anaconda3\lib\site-packages (from qiskit) (0.20.0)

Requirement already satisfied: qiskit-ibmq-provider==0.18.3 in c:\users\admin\anaconda3\lib\site-packages (from qiskit) (0.18.3)
Requirement already satisfied: numpy>=1.16.3 in c:\users\admin\anaconda3\lib\site-packages (from qiskit-aer==0.10.3->qiskit) (1.20.3)
Requirement already satisfied: scipy>=1.0 in c:\users\admin\anaconda3\lib\site-packages (from qiskit-aer==0.10.3->qiskit) (1.7.1)
Requirement already satisfied: requests-ntlm>=1.1.0 in c:\users\admin\anaconda3\lib\site-packages (from qiskit-ibmq-provider==0.18.3->qiskit) (1.1.0)
Requirement already satisfied: urllib3>=1.21.1 in c:\users\admin\anaconda3\lib\site-packages (from qiskit-ibmq-provider==0.18.3->qiskit) (1.26.7)
Requirement already satisfied: requests>=2.19 in c:\users\admin\anaconda3\lib\site-packages (from qiskit-ibmq-provider==0.18.3->qiskit) (2.26.0)
Requirement already satisfied: python-dateutil>=2.8.0 in c:\users\admin\anaconda3\lib\site-packages (from qiskit-ibmq-provider==0.18.3->qiskit) (2.8.2)
Requirement already satisfied: websocket-client>=1.0.1 in c:\users\admin\anaconda3\lib\site-packages (from qiskit-ibmq-provider==0.18.3->qiskit) (1.3.2)
Requirement already satisfied: networkx>=0.8.0 in c:\users\admin\anaconda3\lib\site-packages (from qiskit-ignis==0.7.0->qiskit) (2.6.3)
```

Getting started with Qiskit

```
In [31]: import qiskit
import matplotlib.pyplot
from math import pi
from qiskit.visualization import plot_bloch_vector, plot_bloch_multivector
```

```
In [2]: qiskit.__qiskit_version__

{'qiskit-terra': '0.20.0', 'qiskit-aer': '0.10.3', 'qiskit-ignis': '0.7.0', 'qiskit-ibmq-provider': '0.18.3', 'qiskit-aqua': None,
'qiskit': '0.35.0', 'qiskit-nature': None, 'qiskit-finance': None, 'qiskit-optimization': None, 'qiskit-machine-learning': None}
```

```
In [3]: from qiskit import IBMQ
```

Getting started with Qiskit

```
from qiskit import IBMQ
```

```
IBMQ.save_account(secret_token)
```

Secret ^^

```
IBMQ.load_account()
```


Drawing Bloch sphere using Qiskit

```
plot_bloch_vector([0,1,0], title = "Bloch Sphere")
```

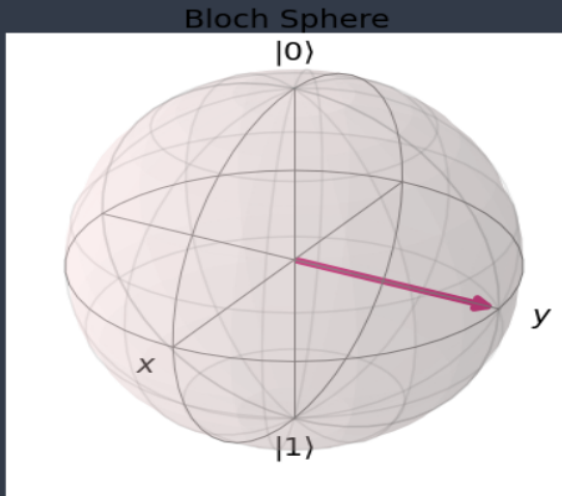


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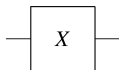
Definition

A quantum gate is a unitary transformation $\mathcal{U} : \mathbb{H}^{\otimes n} \rightarrow \mathbb{H}^{\otimes n}$.
If $n = 1$, we call it unary, and if $n = 2$, we call it binary.

Three representations of a quantum gate: Operator, matrix, and symbol.

Examples of unary transformation : $V : \mathbb{H} \rightarrow \mathbb{H}$, unitary 2×2 matrix.

1. Pauli-X gate (Or Q-Not gate)

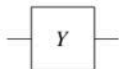


$$X := \sigma_x$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

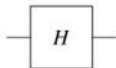
Quantum Circuits : A review

2. Pauli-Y gate



$$Y := \sigma_y \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

3. Hadamard gate



$$H := \frac{\sigma_x + \sigma_z}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

4. Measurement of observable A (this is not a gate).



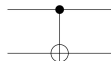
[Not a gate, but a non-unitary transformation of the input-state to an eigenstate of A and delivery of measured value λ]

Quantum Circuits : A review

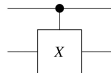
Examples of binary quantum gate : $\mathcal{U} : \mathbb{H}^{\otimes 2} \rightarrow \mathbb{H}^{\otimes 2}$ with basis $\{|0\rangle^2, |1\rangle^2, |2\rangle^2, |3\rangle^2\}$.

1. $\Lambda^1(X)$

Controlled
NOT
(C-NOT)

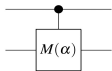


$$\begin{aligned} \Lambda^1(X) &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\ &= |0\rangle\langle 0| \otimes \mathbf{1} + |1\rangle\langle 1| \otimes X \end{aligned}$$

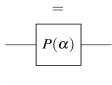


2.

Controlled
phase-
multiplication



$$\begin{aligned} \Lambda^1(M(\alpha)) &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{i\alpha} & 0 \\ 0 & 0 & 0 & e^{i\alpha} \end{pmatrix} \\ &= |0\rangle\langle 0| \otimes \mathbf{1} + |1\rangle\langle 1| \otimes e^{i\alpha} \mathbf{1} \end{aligned}$$



$$P(\alpha) \otimes \mathbf{1}$$

Definition

Let $n, L \in \mathbb{N}$ and let $\mathcal{U}_1, \dots, \mathcal{U}_L \in \mathcal{U}(\mathbb{H}^{\otimes n})$ be quantum n -gates. Then the composition operator $U = \mathcal{U}_L \cdots \mathcal{U}_1$ is called the quantum circuit of depth L .

Remark : There are two other definitions of the quantum circuit. See our previous seminar book on the quantum computation.

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However, before those... let's play with

<https://algassert.com/quirk>

Menu Export Clear Circuit Clear ALL Undo Redo Make Gate Version 2.3

Toolbox

Probes	Displays	Half Turns	Quarter Turns	Eighth Turns	Spinning	Formulaic	Parametrized	Sampling	Parity
$ 0\rangle\langle 0 $ $ 1\rangle\langle 1 $	Density Bloch	Z Swap	S S^{-1}	T T^{-1}	Z^t Z^{-t}	$Z^{f(t)}$ $R_z(f(t))$	$Z^{A/2^n}$ $Z^{-A/2^n}$	Z $Z \otimes 0\rangle$	Z $Z \otimes 0\rangle$
$ 0\rangle\langle 0 $ $ 1\rangle\langle 1 $	Chance Amps	Y	$Y^{1/2}$ $Y^{-1/2}$	$Y^{1/4}$ $Y^{-1/4}$	Y^t Y^{-t}	$Y^{f(t)}$ $R_y(f(t))$	$Y^{A/2^n}$ $Y^{-A/2^n}$	Y $Y \otimes 0\rangle$	Y $Y \otimes 0\rangle$
$ 0\rangle\langle 0 $ $ 1\rangle\langle 1 $		$\oplus$ H	$X^{1/2}$ $X^{-1/2}$	$X^{1/4}$ $X^{-1/4}$	X^t X^{-t}	$X^{f(t)}$ $R_x(f(t))$	$X^{A/2^n}$ $X^{-A/2^n}$	X $X \otimes 0\rangle$	X $X \otimes 0\rangle$

use controls
drag gates onto circuit
outputs change

Local wire states (Chance/Bloch) Final amplitudes

Toolbox2

X/Y Probes	Order	Frequency	Inputs	Arithmetic	Compare	Modular	Scalar	Custom Gates
$ +\rangle\langle + $ $ -\rangle\langle - $	$+ t $ $- t $	QFT $QFT^\dagger$	Input A $A=\#$ default	$+1$ -1	$\oplus A < B$ $\oplus A > B$	$+1 \bmod B$ $-1 \bmod B$	$\dots$ 0	
$ i\rangle\langle i $ $ i\rangle\langle -i $	Reverse	Grad $1/2$ $Grad^{-1/2}$	Input B $B=\#$ default	$+A$ $-A$	$\oplus A \leq B$ $\oplus A \geq B$	$+A \bmod B$ $-A \bmod B$	$-$	
$ i\rangle\langle i $ $ i\rangle\langle -i $		Grad t $Grad^{-t}$	Input R $R=\#$ default	$+AB$ $-AB$	$\oplus A = B$ $\oplus A \neq B$	$\times A \bmod B$ $\times A^{-1} \bmod B$	i $-i$	
				$\times A$ $\times A^{-1}$		$\times B \bmod B$ $\times B^{-1} \bmod B$	$\sqrt{i}$ $\sqrt{-i}$	

Getting started with Cirq

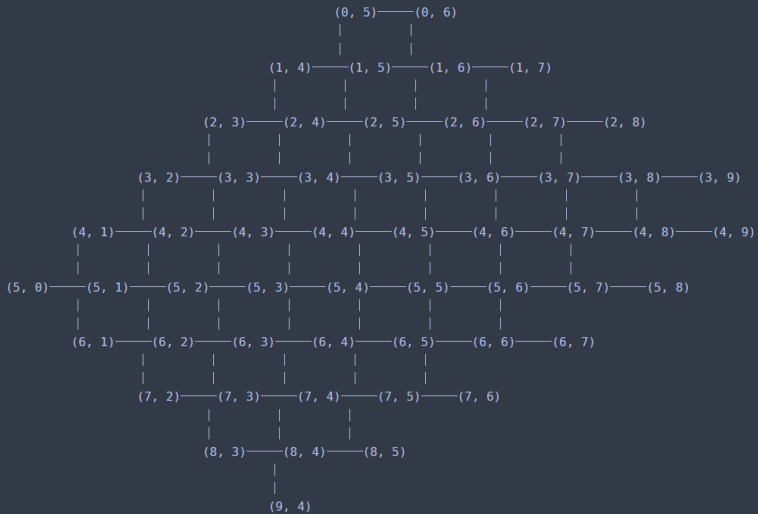
Please see : <https://www.qmunity.tech/tutorials/introduction-to-cirq>

```
In [ ]: pip install cirq
```

```
In [1]: import cirq
```

```
In [2]: import cirq
import cirq_google
```

```
In [3]: print(cirq_google.Sycamore) #Not Bristlecone it is deprecated
```



Creating qubits

```
In [12]: q0=cirq.NamedQubit("q0")
          q1=cirq.NamedQubit("q1")
          print(q0, q1)
          # Line qubits
          q3 = cirq.LineQubit(4)
          print(q3)
          # Creates LineQubit(0), LineQubit(1), LineQubit(2)
          q0, q1, q2 = cirq.LineQubit.range(3)
          print(q0,q1,q2)
          # Grid Qubits can also be referenced individually
          q0_1= cirq.GridQubit(0,1)
          print(q0_1)
          # Create 16 qubits from (0,0) to (7,7)
          qubits = cirq.GridQubit.square(8)
          print(qubits)
```

Creating qubits : Result

```
q0 q1
4
0 1 2
(0, 1)
[cirq.GridQubit(0, 0), cirq.GridQubit(0, 1), cirq.GridQubit(0, 2), cirq.GridQubit(0, 3), cirq.GridQubit(0, 4), cirq.GridQubit(0, 5),
cirq.GridQubit(0, 6), cirq.GridQubit(0, 7), cirq.GridQubit(1, 0), cirq.GridQubit(1, 1), cirq.GridQubit(1, 2), cirq.GridQubit(1, 3),
cirq.GridQubit(1, 4), cirq.GridQubit(1, 5), cirq.GridQubit(1, 6), cirq.GridQubit(1, 7), cirq.GridQubit(2, 0), cirq.GridQubit(2, 1),
cirq.GridQubit(2, 2), cirq.GridQubit(2, 3), cirq.GridQubit(2, 4), cirq.GridQubit(2, 5), cirq.GridQubit(2, 6), cirq.GridQubit(2, 7),
cirq.GridQubit(3, 0), cirq.GridQubit(3, 1), cirq.GridQubit(3, 2), cirq.GridQubit(3, 3), cirq.GridQubit(3, 4), cirq.GridQubit(3, 5),
cirq.GridQubit(3, 6), cirq.GridQubit(3, 7), cirq.GridQubit(4, 0), cirq.GridQubit(4, 1), cirq.GridQubit(4, 2), cirq.GridQubit(4, 3),
cirq.GridQubit(4, 4), cirq.GridQubit(4, 5), cirq.GridQubit(4, 6), cirq.GridQubit(4, 7), cirq.GridQubit(5, 0), cirq.GridQubit(5, 1),
cirq.GridQubit(5, 2), cirq.GridQubit(5, 3), cirq.GridQubit(5, 4), cirq.GridQubit(5, 5), cirq.GridQubit(5, 6), cirq.GridQubit(5, 7),
cirq.GridQubit(6, 0), cirq.GridQubit(6, 1), cirq.GridQubit(6, 2), cirq.GridQubit(6, 3), cirq.GridQubit(6, 4), cirq.GridQubit(6, 5),
cirq.GridQubit(6, 6), cirq.GridQubit(6, 7), cirq.GridQubit(7, 0), cirq.GridQubit(7, 1), cirq.GridQubit(7, 2), cirq.GridQubit(7, 3),
cirq.GridQubit(7, 4), cirq.GridQubit(7, 5), cirq.GridQubit(7, 6), cirq.GridQubit(7, 7)]
```

Creating unitary matrices

```
In [5]: print(cirq.unitary(cirq.X)) #Unitary of the X gate
```

```
[[0.+0.j 1.+0.j]
 [1.+0.j 0.+0.j]]
```

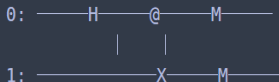
```
In [6]: a, b = cirq.LineQubit.range(2) #Unitary of SWAP operator on two qubits
print(cirq.unitary(cirq.SWAP(a, b)))
```

```
[[1.+0.j 0.+0.j 0.+0.j 0.+0.j]
 [0.+0.j 0.+0.j 1.+0.j 0.+0.j]
 [0.+0.j 1.+0.j 0.+0.j 0.+0.j]
 [0.+0.j 0.+0.j 0.+0.j 1.+0.j]]
```

Creating Circuits

```
In [49]: circuit=cirq.Circuit(cirq.H(q0), cirq.CNOT(q0,q1), cirq.measure(q0,q1))

print(circuit)
```



```
In [53]: simulator=cirq.Simulator()

result=simulator.run(circuit,repitions=30)
print(result)
```

```
0,1=100101101001111111011001111011, 100101101001111111011001111011
```

Creating the Bell state

```
# We will generate the Bell State:
#  $\frac{1}{\sqrt{2}} * (|00\rangle + |11\rangle)$ 
bell_circuit = cirq.Circuit()
a, b = cirq.LineQubit.range(2)
bell_circuit.append(cirq.H(a))
bell_circuit.append(cirq.CNOT(a,b))
print(bell_circuit)

# Initialize Simulator
s=cirq.Simulator()

print('Simulate the circuit:')
results=s.simulate(bell_circuit)
print(results)
print()

# For sampling, we need to add a measurement at the end
bell_circuit.append(cirq.measure(a, b, key='result'))

print('Sample the circuit:')
samples=s.run(bell_circuit, repetitions=1000)
```

Creating the Bell state : The result

0: ———H———@———
 |

1: —————X———

Simulate the circuit:

measurements: (no measurements)

qubits: (cirq.LineQubit(0), cirq.LineQubit(1))

output vector: $0.707|00\rangle + 0.707|11\rangle$

phase:

output vector: $| \rangle$

Sample the circuit:

Qiskit again, Bell state again

```
#quantum circuit to create a Bell state  
bell = QuantumCircuit(2,2)  
print(bell)  
bell.h(0)  
print(bell)  
bell.cx(0,1)  
print(bell)
```

$q_0:$

$q_1:$

$c: 2/$

$q_0:$ $\boxed{H}$

$q_1:$ _____

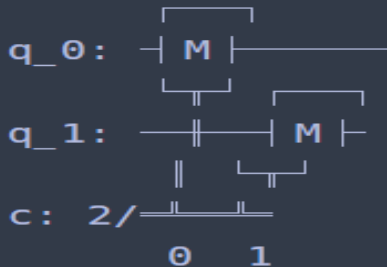
$c: 2/$ =====

$q_0:$ $\boxed{H}$  _____

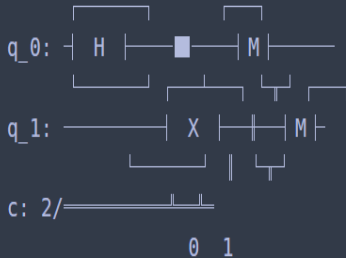
$q_1:$ _____ $\boxed{X}$

$c: 2/$ =====

```
meas = QuantumCircuit(2,2)
meas.measure([0,1],[0,1])
print(meas)
```



```
#execute the quantum circuit
backend = BasicAer.get_backend('qasm_simulator') #the device the run on
circ = bell + meas
print(circ)
```



```
result = execute(circ, backend, shots = 1000).result()  
counts = result.get_counts(circ)  
print(counts)
```

```
{'11': 521, '00': 479}
```

```
from qiskit.visualization import plot_histogram  
plot_histogram(counts)
```

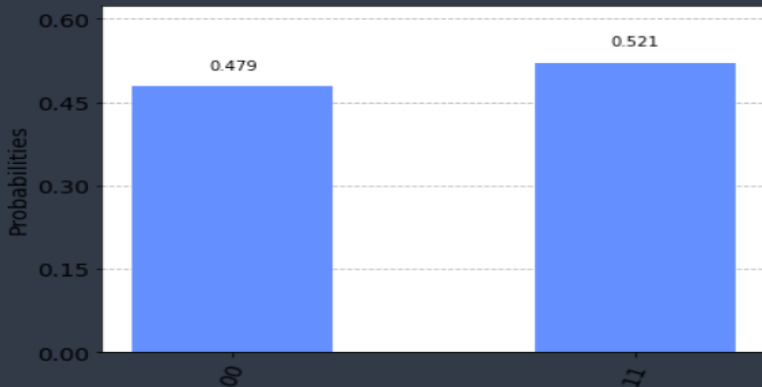


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Von Neumann Entropy

Quantum counterpart of the Shannon entropy

Definition

Given a mixed state $\rho = \sum_{i=1}^n p_i |\psi_i\rangle\langle\psi_i|$, the von Neumann entropy is given by

$$S(\rho) = -\text{Tr}(\rho \log \rho)$$

It is...

1. subadditive, $S(\rho_{12}) \leq S(\rho_1) + S(\rho_2)$.
2. $0 \leq S \leq N = \dim(\mathbb{H})$.

Using Qiskit, one can compute the evolution of the von Neumann entropy of a quantum state.

see you next week :)